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Oefeningenreeks 1B nr. 37.10

$$1 + 2\sqrt{6}i = (x+yi)^2 = x^2 - y^2 + 2xy$$

$$\Rightarrow \begin{cases} x^2 - y^2 = 1 \\ 2xy = 2\sqrt{6} \end{cases}$$

$$\Rightarrow \begin{cases} x^2 + (-y^2) = 1 \\ x^2(-y^2) = -\dfrac{1}{4} \cdot (2\sqrt{6})^2 \end{cases}$$

$$\Rightarrow \begin{cases} x^2 - y^2 = 1 \\ x^2(-y^2) = -6 \end{cases}$$

$$\Rightarrow \lambda^2 - \lambda - 6 = 0$$

$$\Delta = (-1)^2 - 4 \cdot 1 \cdot (-6) = 1 + 4 \cdot 6 = 25$$

$$\lambda_{1,2} = \frac{1 \pm 5}{2}$$

$$\Rightarrow \begin{cases} 3 = x^2 \\ -2 = -y^2 \end{cases}$$

$$x + yi \in \{\sqrt{3} + \sqrt{2}i, -\sqrt{3} - \sqrt{2}i\}$$

References